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The Advantage of Big Team Science

Lessons Learned From Cognitive Science

Vaidis, David C.; Miranda, Jacob F.; Buchanan, Erin M.; Yang, Yu Fang; Kowal, Marta; Schmidt, Kathleen; Topor, Marta; Misiak, Michal; Miller, Rachael; Protzko, John; Gjoneska, Biljana; Miller, Jeremy K.; Exner, Anna; Azevedo, Flavio; Paruzel-Czachura, Mariola; Mushtaq, Faisal; Oliveira, Catia; Wagge, Jordan R.; De Moor, Delphine; Mede, Niels G.; Altschul, Drew M.; Pavlov, Yuri G.; Seetahul, Yashvin; Boucher, Leanne; Doell, Kimberly C.; Visser, Ingmar; Elsherif, Mahmoud M.; Pronizius, Ekaterina

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Methodology and Research Practice

The Advantage of Big Team Science: Lessons Learned From Cognitive Science

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The credibility revolution in psychology and related sciences contributed to the adoption of large-scale research initiatives known as *Big Team Science* (BTS). BTS has made significant advances in addressing issues of replication, statistical power, and diversity through the use of larger samples and more representative cross-cultural data. However, while these collaborations hold great potential, they also introduce unique challenges related to their scale. Drawing on experiences from successful BTS projects, we identified and outlined key strategies for overcoming diversity, volunteering, and capacity challenges. We emphasize the need for clear role definitions, structured and preregistered workflows, centralized project management, and transparent decision documenting to prevent common pitfalls. Ultimately, we call for reflection on the strengths and limitations of BTS to enhance the quality, generalizability, and impact of research across disciplines. This work complements existing BTS guides by offering experientially-grounded, discipline-specific strategies and addressing underexplored logistical, ethical, and epistemological challenges in large-scale collaborations.

Psychology and related sciences have faced a replication crisis, also known as the credibility revolution (Korbmacher et al., 2023; Nosek et al., 2022; Vazire, 2018). This crisis refers to a lack of replication and reliability of various effects found in past studies as a result of limited sample representativeness and small sample sizes. In response, *Big Team Science* (BTS) collaborations emerged. This approach, recognized for its scale, inclusivity, and focus on transparency and replicability, has become a distinct model for advancing the human sciences (Forscher et al., 2023). While various definitions for BTS exist, a common feature is their large scale. This often includes a large number of collaborators (Forscher et al., 2023), substantial data volume (Milkman et al., 2022), along with voluntary contributions and

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bottom-up approaches (Baumgartner et al., 2023). Another definition highlights the importance of community and collaborative motivation overall (Korbmacher et al., 2023). Here, we characterize a BTS project as a large-scale scientific project that is built upon collective effort, undertaken due to the complexity of the scientific issues involved, and facilitated by a distribution of tasks and specialties within the team.

BTS initiatives have extended beyond research on human adults' cognition and behavior, encompassing studies across development (Frank et al., 2017; Hall et al., 2018; Moshontz et al., 2018); species (Alessandroni et al., 2024, 2025); fundamental topics involving core theories (e.g. Tierney et al., 2020, 2021; Vaidis et al., 2024), tools (Botvinik-Nezer et al., 2020; Coretta et al., 2023; Delios et al., 2022; Pavlov et al., 2021), ecological (Gould et al., 2025) and pedagogical research (Mahak et al., 2024; Pownall et al., 2023, 2024; Terry et al., 2023); as well as applied research for climate change and science communication (Cologna et al., 2025; Vlasceanu et al., 2024). Leveraging the strengths of the community, BTS projects have produced high-impact publications demonstrating their ability to tackle key challenges related to replication and the generalization of effects (Ebersole et al., 2016; Klein et al., 2018; Open Science Collaboration, 2015). Despite these strengths, BTS also faces challenges. The goal of the present paper is to identify strategies to enhance the efficiency and impact of collaborative research as BTS continues to evolve.

A growing number of frameworks and practical guides have been developed to support the development and implementation of BTS, including contributions that emphasize structural (Korbmacher et al., 2023), philosophical (Forscher et al., 2023), and procedural (Baumgartner et al., 2023; Coles et al., 2023; Moshontz et al., 2018, 2021) aspects of collaborative research. While these works provide essential foundations for the field, the current manuscript offers a distinct and complementary contribution. First, it uses a problem-solution approach. This approach is based on data gathered from many experienced BTS managers to identify and prioritize challenges according to how often they occur and how deleterious they are. Second, this work specifically addresses cognitive science, providing detailed advice on how to navigate diverse disciplinary perspectives and disagreements. Third, it covers pragmatic operational topics such as coordinating cross-cultural ethics approvals, managing lead team transitions, ensuring project sustainability, funding, organizing meetings across different time zones, and responsible AI use. These recommendations provide a granular and actionable framework for enhancing the success and sustainability of BTS efforts.

Assessing the Challenges in Managing Big Teams

The ideas presented in this paper were elaborated during two hackathons (i.e., collective work sessions; Vaidis et al., 2022, 2023) and are grounded in the collective experience of individuals[1] who have held key leadership and coordination roles in BTS projects. Together, we proposed concrete improvements, and reflected critically on lessons learned across diverse project contexts.

Step 1: Hackathon 1

During an initial 90-minute hackathon session, 39 leaders of various BTS initiatives defined and documented a list of about 100 most common issues and barriers they encountered, anticipated, or were concerned about during their project execution.

Step 2: Systematic Assessment of Challenges

A smaller team of contributors summarized the identified issues by merging duplicates and selecting broader, more general terms. Based on this refined list, they developed a survey to adopt a more systematic and structured approach. A total of 75 proposed challenges, each with a short definition, were included in the survey and were assessed on two dimensions using 4-point Likert scales: *frequency* (1-extremely rare, 2-not very often, 3-often, 4-extremely common) and *severity* (1-not severe at all, 2-slightly severe, 3-moderately severe, 4-extremely severe; full materials are accessible on https://osf.io/fb8gt/). Both initial and new contributors were invited to complete the survey to support the quantifiable categorization of each challenge.

To build on relevant experience, responses from contributors without BTS management background (i.e., those not serving as lead investigators, project managers, or executive-level team members) were excluded from the analysis. Ultimately, 32 responses from experienced BTS project managers were obtained.

Intermediate Step: Results from the Survey

From the initial 75 proposed challenges, and based on the survey responses, we reduced the number of challenges to emphasize distinctiveness and relevance. For clarity and conciseness, conceptually similar challenges showing little-to-no variation in ratings were merged (e.g., *Contributor Turn-Over* and *Slow Staff Turnover*). Similarly, trivial challenges (i.e., assessed as both infrequent and not severe at all) were excluded from the report. This resulted in 61 relevant challenges (see Figures 1-3 and Appendix A), along with their potential impact on project success in terms of frequency and severity. Given the descriptive nature of the

1 The majority of contributors are current authors of the paper. A few others were involved only in the hackathons and survey but declined authorship, considering their contributions insufficient. These individuals are acknowledged in the authors' notes.

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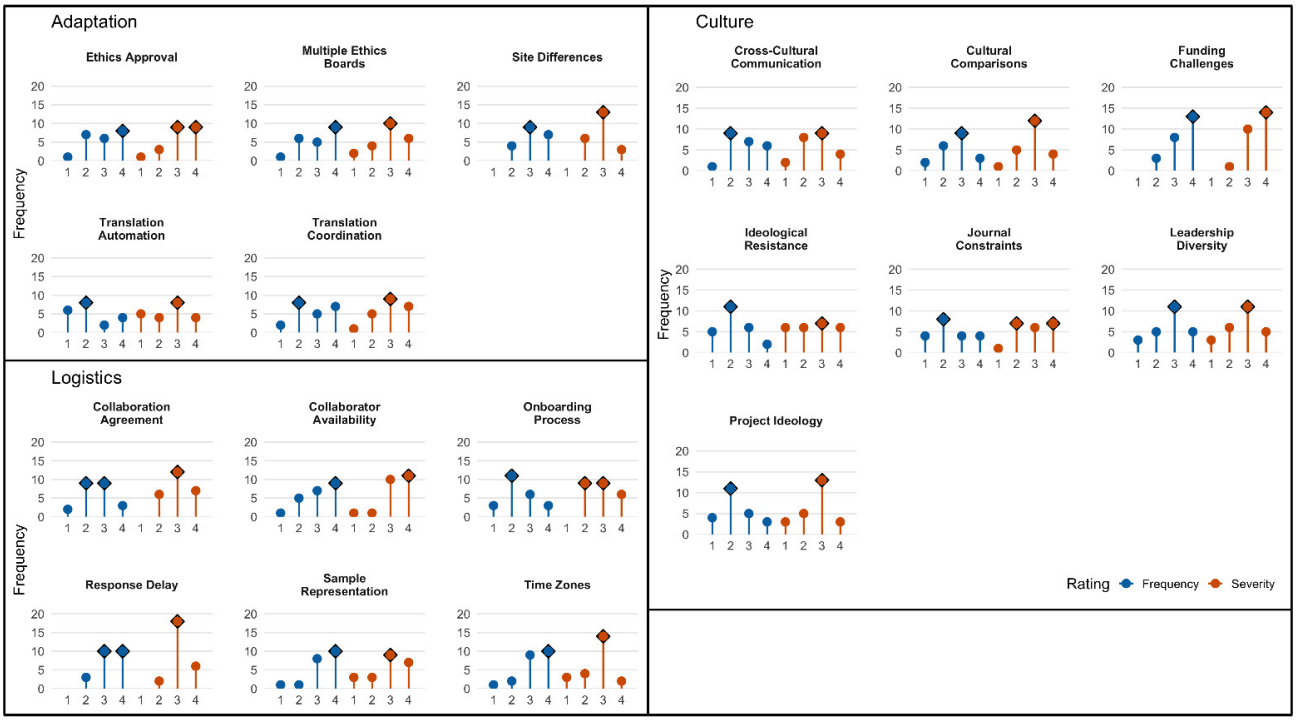


Figure 1. Diversity Challenges in Terms of Frequency and Severity

Note. Lollipop charts showing the distribution of frequency (blue) and severity (orange) scores for barriers in three categories: Adaptation, Culture, and Logistics, with scores ranging from 1 (extremely rare / not severe) to 4 (extremely common / severe). Modes are marked with diamonds. See Appendix A for items' description.

data, no inferential statistical analyses were conducted. Instead, our aim is to offer a practical overview to help BTS newcomers anticipate common obstacles and identify the most critical challenges. To facilitate this identification, the challenges were grouped into three overarching categories: (a) *Diversity*, (b) *Volunteering*, and (c) *Capacity*, each with respective sub-domains. Below, we highlight the challenges that showed the highest agreement among respondents and the least variation in ratings, indicating broad consensus on their importance.

Most Relevant Diversity-Related Issues

Diversity-related challenges were summarized into 18 issues and subcategorized into three areas: *Adaptation*, *Culture*, and *Logistics* (Figure 1). *Adaptation* involves navigating diverse local restrictions and languages (5 items). *Culture* refers to differences in values, norms, and research practices across regions or disciplines, which can sometimes lead to misunderstandings or tensions within the team (7 items). *Logistics* addresses the operational impacts of diversity on human workflows and coordination (6 items).

Among diversity challenges, several elements reached a consensus regarding their potential to severely impact a project and their likelihood of emerging during a BTS project. *Ethics Approval* ($Mode_{Frequency}$ = 4, $Mode_{Severity}$ = 3-4) across multiple data collection sites can be difficult to obtain and may expose BTS projects to various issues depending on local rules (see *Legal and Ethical Considerations*). This difficulty could lead to labs abandoning the project, an incapacity to contribute, or even legal concerns. *Funding Challenges* ($Mode_{Frequency}$ = 4, $Mode_{Severity}$ = 4) emerged as the most frequent and severe challenge. It raises concerns about securing project funding, which directly impacts project prioritization and individual engagement. Additionally, legal complications may arise when labs use funds obtained by a lead team in another country. Ultimately, inadequate funding can cause labs to withdraw from a project—or worse, prevent the project from materializing altogether. Finally, *Collaborator Availability* ($Mode_{Frequency}$ = 4, $Mode_{Severity}$ = 4) and *Response Delay* ($Mode_{Frequency}$ = 3-4, $Mode_{Severity}$ = 3) also emerged as significant constraints. Variability in collaborators' schedules and time commitments can disrupt timelines and affect overall project coordination. Similarly, collaborator responsiveness—the timeliness and effectiveness of communication—plays a critical role that can slow project progress.

Most Relevant Volunteering-Related Issues

Volunteering-related challenges were numerous and summarized into 30 issues, subcategorized into four areas: *Distribution*, *Interpersonal relationship*, *Sustainability*, and *Involvement* (Figure 2). *Distribution* refers to the organizational structure, labor division, empowerment and, more globally, management (7 items). *Interpersonal relationship* refers to the interactions between collaborators and the consequences in terms of management (3 items). *Sustainability* refers to the continuation of the project in the long term and the risk of desertion from both collaborators and the lead team (6 items). Finally, *involvement* refers to the underlying motivational aspects of volunteering and their consequences in terms of (dis)engagement (14 items).

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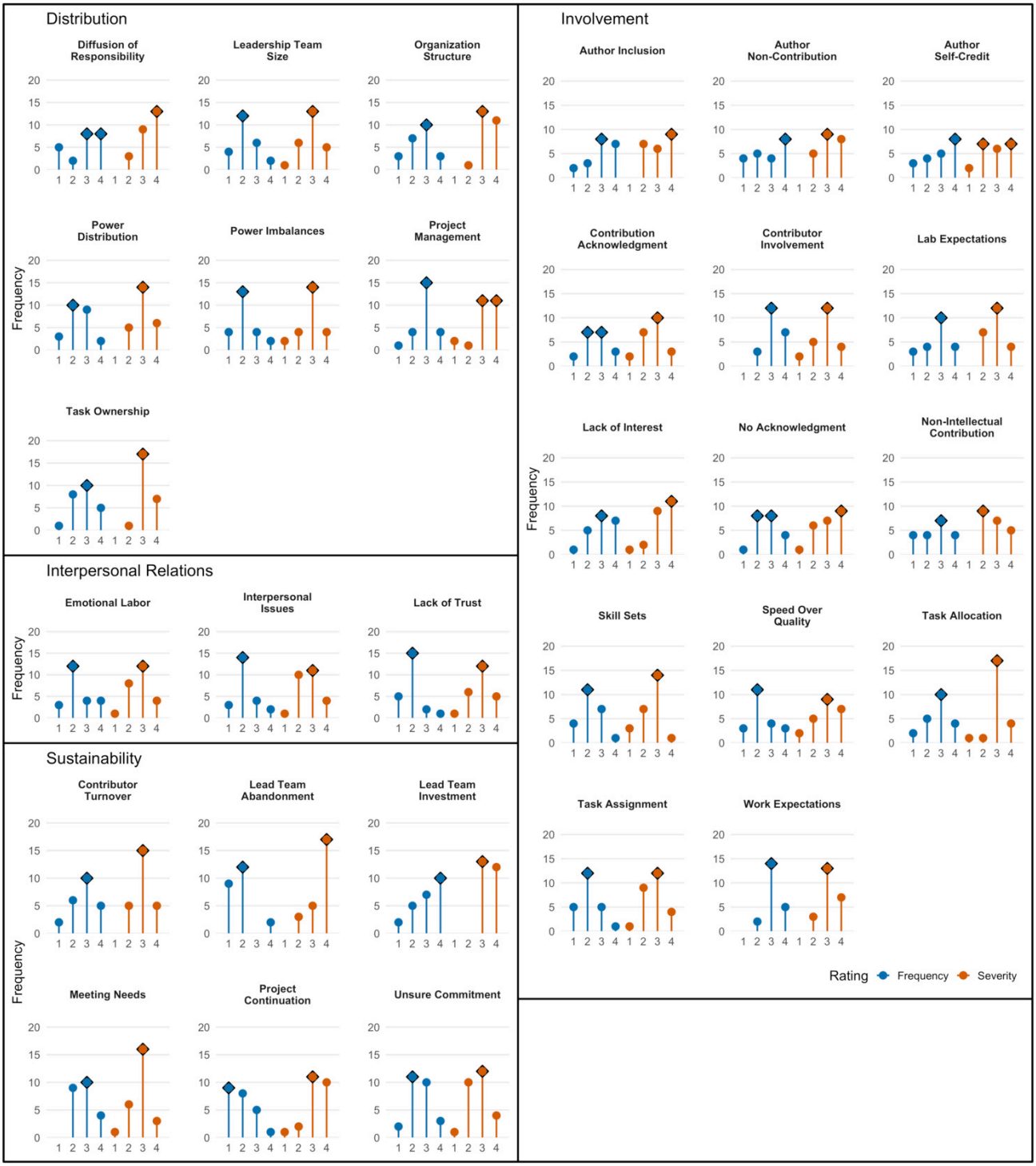


Figure 2. Volunteering Challenges in Terms of Frequency and Severity

Note. Lollipop charts showing the distribution of frequency (blue) and severity (orange) scores for barriers in four categories: Distribution, Interpersonal Relations, Sustainability, and Involvement, with scores ranging from 1 (extremely rare / not severe) to 4 (extremely common / severe). Modes are marked with diamonds. See Appendix A for items' description.

Similar to the previous section, several elements reached consensus among the identified volunteering challenges. *Diffusion of Responsibility* ($Mode_{Frequency}$ = 3-4, $Mode_{Severity}$ = 4) was the most frequent and severe challenge. In the same vein, *Work Expectations* ($Mode_{Frequency}$ = 3, $Mode_{Severity}$ = 3) are often perceived by external observers as lower when a project involves a large number of collaborators. While involving numerous collaborators may seem an effective way to distribute tasks and reduce individual workload, it can also dilute responsibility and weaken commitment. Reduced accountability may undermine motivation and engagement, compromising data quality and slowing overall project progress. Moreover, scaling up a project does not necessarily lessen individual effort; instead, it often increases demands related to translation, compliance with established procedures, and coordination to maintain timelines. Without adequate communication and management, large-scale collaborations risk becoming inefficient. These

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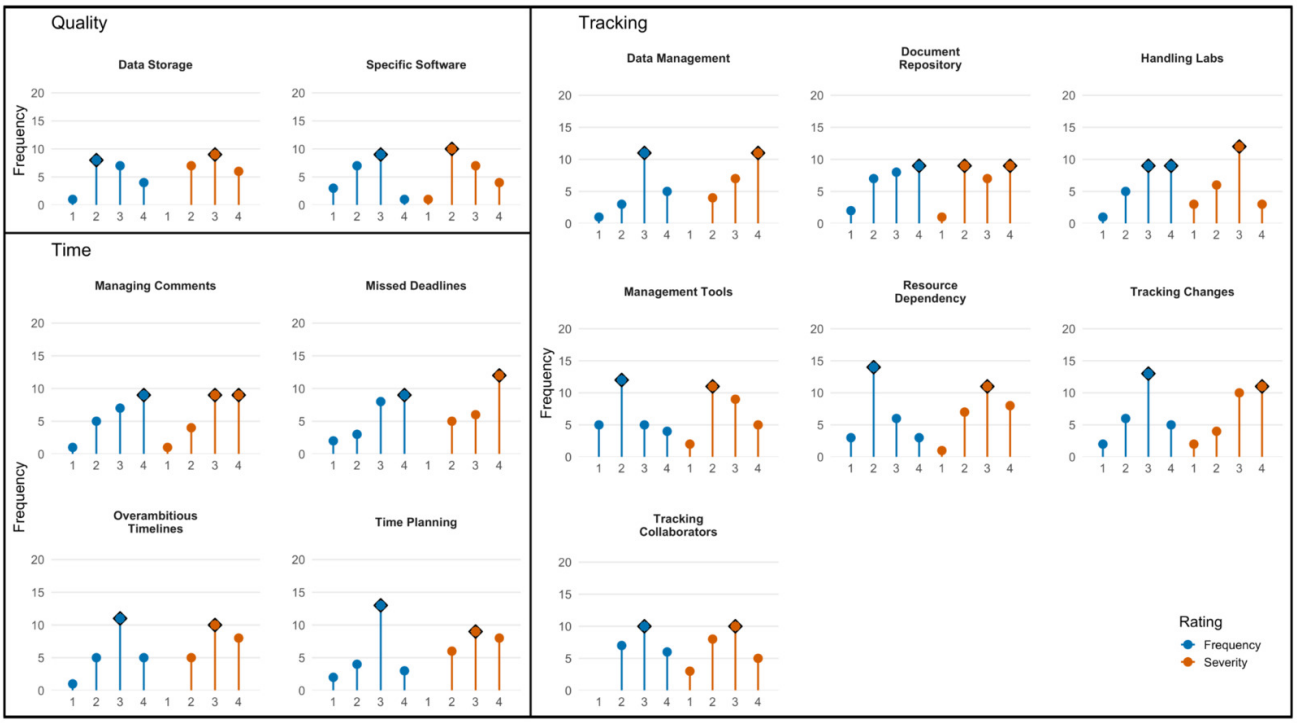


Figure 3. Capacity Challenges in Terms of Frequency and Severity

Note. Lollipop charts showing the distribution of frequency (blue) and severity (orange) scores for barriers in three categories: Quality, Time, and Tracking, with scores ranging from 1 (extremely rare / not severe) to 4 (extremely common / severe). Modes are marked with diamonds. See Appendix A for items' description.

challenges are discussed in detail in the sections on volunteering challenges, which outline both the importance of and strategies for effectively managing volunteer contributions in the BTS context. Finally, *Lead Team Abandonment* has been identified as a major threat to BTS projects ($Mode_{Frequency}$ = 2, $Mode_{Severity}$ = 4). Although it is not reported as frequent, this perception may be partially explained by a survivor bias—contributors were specifically those who successfully led their projects to completion. Yet, the existence of unfinished projects, abandoned BTS initiatives, or unresponsive lead team members is a reality well known within the community.

Most Relevant Capacity-Related Issues

Capacity-related issues refer to limited capacity for humans to organize, compute and manage information or tasks. There were 13 issues, subcategorized into three areas: *Quality*, *Time*, and *Tracking* (Figure 3). Quality refers to the data storage practices and the use of specific software that may support or hinder the achievement of high-quality outcomes (2 items). *Time* directly refers to the management of delays (4 items). *Tracking category* (7 items) refers to the necessity to mark and record all relevant aspects of the project to allow for adjustments and decision-making.

Several items found a consensus in terms of both frequency and severity, highlighting key capacity-related bottlenecks. Among the most critical were *Missed Deadlines* and *Tracking Changes*, both consistently rated as highly frequent and severe ($Mode_{Frequency}$ = 4, $Mode_{Severity}$ = 4). Missed deadlines refers to the difficulty to make sure that every contributor did the task on time, which requires monitoring of each lab and each task. Tracking changes illustrates the cognitive load required to handle multiple streams of information. While often overlooked, effective tracking—whether of collaborators, documents, or updates—emerged as a central challenge for sustaining long-term progress. These findings underscore the need for robust tracking systems and streamlined workflows to prevent fragmentation and inefficiencies in BTS project execution. These issues and solutions are developed in the eponymous section.

Step 3: Hackathon 2

After a second hackathon (following the same structure as the first) and subsequent refinement, three primary challenges were identified, which we elaborate on in the current manuscript: *addressing diversity*, *fostering volunteering*, and *reinforcing capacity*. Each challenge reflects broader concerns, such as cultural and international adaptation, maintaining effort and motivation, and overcoming human limitations. In the following sections, we discuss how each of these issues may threaten the success of a project and suggest practical strategies to mitigate them.

Strategic Reflections for BTS Success

Diversity Challenges: Cultural Adaptation and International Management

BTS projects are inherently international, encompassing collaborators with diverse global locations, organizations, languages, time zones, and statuses (e.g., income, institu-

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Table 1. Summary of Recommendations for Diversity Challenges

Challenges	Recommendations
Epistemological Tensions	• Make epistemological assumptions explicit early • Use structured collaboration (e.g., adversarial collaboration) to manage disagreements • Set a conflict pathway: attempt consensus-building first and decide what's next if unresolved
Legal and Ethical Considerations	• Establish a workgroup to ensure ethical and legal compliance with local requirements (e.g., restrictions on data types and necessary consent forms) as early as possible • Leverage local collaborators or data collectors to provide overviews of local legal frameworks and institutional policies • Design studies iteratively with ethics constraints in mind
Material Adaptation	• Set up translation workgroup and/or assign one responsible supervisor per language • Establish a clear translation procedure and validation • Issue translation certificates to formally recognize translators' contributions
Meetings in a Global Setting	• Plan meetings in advance, circulate agendas, share minutes promptly • Use Coordinated Universal Time (UTC) for scheduling meetings with multi-time zone teams • Organize time zone hubs and appoint rotating leads to bridge them, ensuring effective information relay • Standardize logistics and enable asynchronous communication
Communication in a Multicultural World	• Acknowledge different cultural communication styles and take time to chat for getting more awareness for these differences • Recognize implicit cultural assumptions and biases to promote equitable collaboration • Use clear, simple language to avoid misunderstandings, rely on *Universal Design for Learning* to improve understanding and inclusion.

tional support, and workload allocation for research). To ensure research is generalizable and meaningful, it is crucial that collaborators with varied theoretical and cultural backgrounds be included in the data collection and study design—ideally on the lead or methods teams. Table 1 summarizes recommendations for diversity challenges.

Epistemological Tension

Integrating researchers with different backgrounds into a productive BTS consortium presents unique challenges, as BTS project managers must navigate not only cultural differences but also disagreements about knowledge and methodological traditions. It is rare for all members of a BTS project to agree on every foundational assumption (see Sulik et al., 2025). Indeed, disciplinary capture (Brister, 2016) refers to the idea that different, yet rational, individuals can hold different assumptions about how a concept is operationalised, which approaches and methods are valid, and which facts, causes and research goals are relevant.

Two main approaches can help navigate, though not fully resolve, epistemological tensions: the disciplinary perspectives framework (van Baalen & Boon, 2024) and the adversarial collaboration (Kahneman & Klein, 2009). The disciplinary perspectives framework helps alleviate discipline capture by encouraging individuals to explicitly discuss their epistemological assumptions to increase understanding of why they may raise certain objections. The framework consists of eight questions that research teams are recommended to make explicit at the start of the collaboration. Questions such as, "*What is the overarching goal of the (disciplinary) professional or research practice?*" or "*Which methodology and (technological) instruments to explain or investigate the phenomena of interest are typically used in this discipline?*" can help elucidate differences that were not previously acknowledged.

Adversarial collaboration involves scientists with competing views, assumptions and theories working together to collectively test their hypotheses. Importantly, this approach requires a baseline of shared methodological ground; it is most effective when disagreements concern empirical predictions rather than what constitutes valid evidence or research questions. This process requires all perspectives to agree, upfront and to the extent possible, on the basic facts, appropriate methods, analyses, and design to address the shared question of interest. All sides build mutual trust and commit to interpret the data from their respective viewpoints afterward (see the following examples: Bena et al., 2022; Coles et al., 2022; Isch et al., 2025; Martel et al., 2024). This allows each side to critically assess how the data fit their narrative, with an opportunity to share dissenting opinions. While it does not resolve all issues, this approach helps make each researcher's implicit assumptions explicit. This encourages researchers to move forward with a pragmatic consensus on theoretical and methodological design, even while maintaining their distinct views on the nature of the knowledge produced.

Navigating these tensions can also be achieved through various means, such as collaborative workshops within teams in charge of different parts of the project that aim to develop consensus (Reilly et al., 2024) or convergence on competing epistemologies. Alternatively, top-down decisions, where project leaders resolve conflicting theoretical or methodological approaches, can be employed as a last resort to settle disagreements before the collaboration is ended or significantly altered.

Finally, it is important to accept that not every disagreement can or should be resolved. Competing paradigms may

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not merely offer different perspectives on the same problem, but may rest on fundamentally incompatible assumptions about the phenomenon's nature, its representation, and its explanation. Researchers with different meta-assumptions may therefore conceive of topics in radically different ways, making integration difficult or even impossible. In such cases, where agreement or collaboration is not feasible, researchers should be transparent about these differences and acknowledge them as limitations, especially if a group of conceptually-aligned researchers decides to pursue a BTS project.

Legal and Ethical Considerations

Legal regulations, both national or international, define the rights and constraints of the researchers. International norms often stem from treaties (e.g., the Declaration of Helsinki), which are ratified by countries. While no single global law applies universally, national and institutional regulations establish the framework for ethical and legal research conduct. However, these local regulations are variable, differing not only between countries but also across institutions. This variability can raise numerous potential issues related to aspects such as data collection, project ownership, and compliance with local ethical laws. In this complex environment, the lead team, with support from local labs, must ensure the project aligns with the local regulations, while upholding respect for universal rights across all settings as the ultimate goal of all ethics procedures.

Some collaborators may be from institutions or countries that lack ethical oversight of research or where reliance on a central ethical approval is permissible. This means that participating labs, including those collecting data, might not require an ethical board review from their own institution or country. In such cases, we recommend obtaining a letter of attestation from the lab's administration, preferably on university letterhead. As this can be a lengthy process for some labs, identifying such labs early allows sufficient time to secure the document before the project reaches its ethics stage. Collected attestation shall then be stored in the central project repository.

Other institutions may demand extensive detail for their ethics review applications. The time required to process an application or an amendment also varies between countries and institutions, ranging from weeks to several months. Additionally, ethics review applications may involve fees that can be prohibitive for collaborator participation. Finally, different ethical standards may lead to varying categorization of sensitive data, limiting the type of information that can be collected and shared (e.g., *General Data Protection Regulation*, GDPR, in the European Union).

The recommended approach to address these and other challenges is to establish an ethics workgroup with a dedicated ethics coordinator for the entire project. This ethics coordinator's role would be to acquire timeline estimates, recommendations from participating labs, and ultimate requirements. Members of the ethics workgroup should represent diverse countries and/or institutions, given the wide variation in ethics requirements by location.

To mitigate delays in the ethics approval process, we highly recommend thorough documentation outlining the ethics review application requirements for each participating lab. This should include time estimation for experimental procedure, relevant data restrictions, and other site-specific constraints (e.g., fees). In our experience, sharing of templates and materials has proven useful in supporting individual labs as they prepare their applications for institutional and ethics board reviews. Such a structured process of an ethics workgroup would better support participating labs in securing ethics board approval.

Further, an ethics workgroup can help identify the types of data that can or cannot be collected without specific restrictions. For example, data on participant characteristics, such as gender identity, ethnic background or postal code can have different legal status depending on the country, institution, and field of research. In the USA, postal codes are considered protected health information and generally cannot be shared in medical fields. By contrast, such data can be collected by communications or psychological researchers. Furthermore, labs in European countries subject to the GDPR might not be able to upload video recordings of participants, which poses challenges for studies using facial or eye-tracking technology (e.g., Menéndez González & Bozkir, 2024). Understanding such local restrictions can significantly aid the initial study's design and programming.

A further complication to the matter is the order of operation: study design, ethics approval, and translation are interdependent. We recommend drafting the study design within the lead team first, while considering general ethical requirements. Then, seek feedback from other labs about local ethical constraints and translation needs before finalizing the study protocol. This iterative process ensures that the final design accommodates both scientific and local ethical considerations before translation and submission for ethics approval proceed in each country.

Material Adaptation to Language and Culture

A crucial aspect of material adaptation is translating the original material into the language of the participating labs (so-called, forward translation)[2]. While some research materials (e.g., questionnaires) may have pre-existing validated versions in multiple languages, it is improbable that all languages and cultures are adequately represented. Even when translations are available, it is not guaranteed that these versions demonstrate measurement invariance, meaning the construct being measured may not be inter-

2 For one example of a translation team approach in a BTS project with 44 languages and dialects, see the Psychological Science Accelerator methodological description under the heading "Translation" (https://osf.io/shn5r).

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preted in the same way across different linguistic and cultural groups (Oshiro et al., 2024).

For this reason, a clear strategy for translating study materials or working with existing translated materials should be integrated into the project's plan from the outset. One approach is to establish a translation workgroup and/or language-specific coordinators tasked with recruiting external translators, overseeing the translation process, and distributing adapted study materials. Alternatively, this responsibility can be delegated to the participating labs themselves. In such cases, the lead team should provide clear guidelines for translation and adaptation to ensure consistency across all participating labs. While the existing established translation and adaptation guidelines (Beaton et al., 2000) ensure a rigorous translation process, they may overlook important contextual details that are specific to each individual study (Epstein et al., 2015) and each individual language (Kowal, 2024). Therefore, it is crucial that translators have a clear understanding of the study's objectives, including details about the rationale behind the methods. They also should document any uncertainties, which are then reviewed by at least one additional person before the materials are finalized. Additionally, we recommend requiring back translations (i.e., the process of translating a previously translated text back to its source language) that are checked by research topic experts for equivalent meaning, especially in projects with subtle manipulations or linguistic nuances in the materials.

Crucially, the lead team should ensure that all translators are appropriately credited for their contributions, as their meticulous work that is so vital for the global reach and success of these projects, tends to be often overlooked (see also Volunteer Challenges). A translation certificate can serve as public recognition, but both the certificate and the individual's role must be visible to be meaningful. We recommend issuing one certificate for language-specific coordinators while listing individual translators on a separate certificate, detailing the specific materials translated and the languages involved. This approach celebrates both individual and collective achievements, while ensuring accountability: if errors arise, this designated individual is responsible for addressing them. This dual aspect of public recognition and responsibility aligns seamlessly with frameworks like MeRIT (Nakagawa et al., 2023).

Meeting in a Global Setting

In BTS projects spanning multiple time zones, scheduling challenges are common due to varying working hours, daylight saving time adjustments, and other constraints. Meeting schedules should be planned well in advance, using Coordinated Universal Time (UTC) to prevent time-related confusion. Meetings are recommended to be concise, adhere to strict schedules, and be prepared with a detailed agenda shared with collaborators beforehand. Dedicating time for personal introductions and cultural exchanges enriches the BTS experience, fostering unity among team members from diverse backgrounds. This not only strengthens interpersonal relationships but also imbues the project with a wealth of cultural perspectives and insights (see also Volunteering Challenges).

Despite efforts, accommodating the schedules of many collaborators is rarely feasible. It may be necessary to organize work into hubs based on daytime working hours, with adjacent continents having overlapping schedules (see Supplementary Materials). We recommend appointing at least one rotating lead team member to join subgroup meetings across time zones, ensuring effective information sharing. Everyday logistics (e.g., onboarding, document versioning, meeting minutes) can be streamlined through templated forms and free collaboration platforms such as OSF (https://osf.io/), keeping the extra burden per person minimal. These strategies bolster asynchronous communication, ensure robust record-keeping, and maintain clarity and continuity regarding decisions, deadlines, and responsibilities (see also Capacity Challenges).

Communication in a Multicultural World

Taking perspective in communication is crucial, especially in our multicultural world. People contribute to BTS with diverse backgrounds, values, cultural communication styles, and varied cognitive and linguistic profiles. These differences can create challenges in understanding one another, not due to any individual's limitation but because effective communication requires bridging distinct perspectives (e.g., Elsherif, 2025; Milton, 2012). Consequently, both verbal and written communication can be fraught with biases and misunderstandings, especially in BTS projects involving diverse participating labs from various locations, each with different norms.

Common biases like false-consensus (Marks & Miller, 1987) can cause a project lead to assume their norms are universal or self-evident, often without questioning their applicability in other contexts. For instance, the lead team might assume their specific regulatory situation, drawing on previously discussed legal frameworks, is universally applicable, overlooking potential variations or equivalents in other countries. Another example is the notable diversity in preferred communication channels across regions; some collaborators may favor direct messaging platforms for work-related exchanges, while others reserve them for private matters, the differences that can potentially impact collaboration dynamics.

To counter this, the lead team should cultivate cultural awareness, recognizing social differences and the unequal distribution of resources across regions. This ensures project requirements and suggestions are feasible across different contexts. This may require posing basic but essential questions, a task that falls naturally within a coordinator's role. When raised from a leadership position, such questions tend to be better received and can help preempt misunderstandings or tensions during meetings. Ultimately, successful communication within multicultural teams depends on a conscious effort to decentralize the lead team's assumptions and adopt a more inclusive perspective. This shift is primarily enacted through language and the manner in which the lead team engages with its collaborators across various cultural and national settings.

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BTS projects, frequently conducted by institutions in the Global North, often use English as a *lingua franca*. This presents invisible barriers for non-native speakers (Amano et al., 2023; Puthillam et al., 2024). When communicating with a global audience, certain linguistic elements pose comprehension challenges, necessitating more explicit articulations. Figures, numbers, and proper names are often difficult to process aurally due to their precise, culturally specific pronunciations. While it might seem redundant, explicitly stating what these figures, numbers, or names actually represent or who they belong to can significantly enhance inclusivity and disambiguate speech. Similarly, using plain language and avoiding jargon and acronyms is crucial for ensuring comprehension across diverse expertise levels. Additionally, idioms present a significant hurdle, as their meanings are non-literal and deeply tied to cultural contexts. While adopting others' perspectives and expending more cognitive effort is necessary for effective global communication, it is important to remember that understanding a foreign language already demands additional resources from the audience.

Ultimately, being understood is a fundamental and invaluable goal in communication. We recommend applying Universal Design for Learning (UDL) principles to provide information in multiple formats (text, audio, video) in order to enhance inclusivity (Puthillam et al., 2024; Rao et al., 2023). UDL's 3.0 Guidelines emphasize three core principles (Rao et al., 2023). First, multiple means of engagement aim to stimulate interest and motivation through autonomy and clear task relevance. In BTS, this could mean allowing asynchronous collaboration, accommodating diverse personal circumstances by not mandating meeting attendance when unnecessary. Second, multiple means of representation focus on delivering accessible information. Additionally, information should be presented multimodally (e.g., closed caption, infographics) to enhance engagement and accessibility. Thirdly, multiple means of action and expression promote inclusive participation through flexible roles and diverse contribution modalities (e.g., writing, data analysis, team coordination). We encourage BTS project managers to explore UDL's wide-ranging applications, including normalizing reasonable adjustments, providing software training, and ensuring equitable leadership opportunities for underrepresented scholars.

Volunteering Challenges: Managing Sustainable Efforts

For BTS projects, collective work is essential, yet most are carried out by uncompensated volunteers with limited resources. Therefore, maintaining motivation and preventing overburdening of both collaborators and lead teams is crucial for the project's success. This section explores four key strategies shown to effectively reduce the impact of such challenges in previous projects (Table 2).

Adoption of Sustainable Project Management

A major challenge for BTS is the substantial workload, as a BTS project can take several years to complete. We recommend dividing tasks according to individual expertise, allowing BTS collaborators to contribute in their areas of specialization (e.g., data analysis, visualization or data collection). To prevent delays when collaborators or the lead team become overwhelmed, effective delegation of tasks and fair distribution of the workload are necessary.

Regular meetings with participating labs and workgroups have been proven to be valuable for fostering engagement, setting expectations, reducing ambiguity, and building a sense of community. They also provide opportunities to clarify tasks, offer timeline reminders, and allow collaborators to ask questions.

In addition, establishing a centralized, easily accessible online hub, such as an OSF project page or a dedicated website, can significantly enhance coordination (see also Capacity Challenges). A central location for updates, documents, and communication reduces reliance on email (which can become fragmented over time) and ensures that team members always know where to find essential information, even as personnel or roles evolve throughout the project's lifespan.

Distribution, not Diffusion, of Responsibilities

The success of BTS often hinges on empowering collaborators. Rather than relying solely on a few individuals (e.g., the lead team), these projects thrive through a distributed collaborative process, where individuals can genuinely influence the final outcome. Yet, while reducing the authority of the lead team, it is crucial to provide clear goals and accountability (Zhu et al., 2018). The challenge lies in distributing power without diluting individual responsibility and agency (i.e., diffusion of responsibility; see Simms & Nichols, 2014 for review), as ambiguity regarding key project responsibilities can lead to poor decision-making. To mitigate this risk, it is necessary to assign clear task responsibilities to collaborators, with some taking on coordination roles for specific work packages. If they cannot complete the task independently, having backups or co-pilots is a valuable practice. Collaborators need to feel confident in their abilities, meaning they are competent, recognized, and willing to take on responsibilities. Instead of relying solely on top-down assignment, it can be beneficial to implement holacracy, where responsibilities are nominated based on acknowledgment from a workgroup they are part of. Whether using a top-down or bottom-up approach, the selection process should be fair, but efficiency and reliability remain crucial. The lead team should avoid working with inefficient or error-prone collaborators: If domain-specific expertise is required, requesting proof of competence (e.g., previous experience, translation certificate) is advisable, and any necessary training should take less time than completing the task internally.

Regardless of the chosen approach, the formal organizational structure and decision authority must be explicitly defined and clear to all members. Ambiguity in these areas leads to confusion over responsibility for key project decisions. Establishing clear roles and agreeing upon the decision-making process (e.g., Figure 4) is essential, especially

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Table 2. Summary of Recommendations for Volunteering Challenges

Challenges	Recommendations
Adoption of Sustainable Project Management	• Delegate tasks fairly and empower skilled collaborators to choose roles they excel in (e.g., data analysis, visualization) • Hold regular all-hands/town-hall meetings to build engagement • Use a centralized online hub to coordinate work and reduce email overload
Distribution, not Diffusion, of Responsibilities	• Define roles and responsibilities (e.g., project manager, coordinator), if appropriate by nominations of (co)pilots (e.g., holacracy) • Assign deliverables to collaborators with transparent accountability • Define formal organizational structure and decision-making process • Set up a policy document (e.g., a collaborator agreement) that defines decision-making processes, responsibilities, expectations, and consequences
Fairness but not Fuzziness with the Contributors' Lack of Involvement	• Outline expectations in a contributor agreement that defines individual roles, timelines, and communication channels • Remind collaborators of the importance of research integrity, data quality, and long-term impact • Report contributions transparently using systems like CRediT taxonomy • The lead team reserves the right to remove individuals should they fail to meet the requirements.
Covering Lead Team Transition	• Define ownership of project components. • Carefully select (co)leads and ensure shared access to core documents (data, contributors, etc.) • Support (co)leads through dedicated regular meetings.

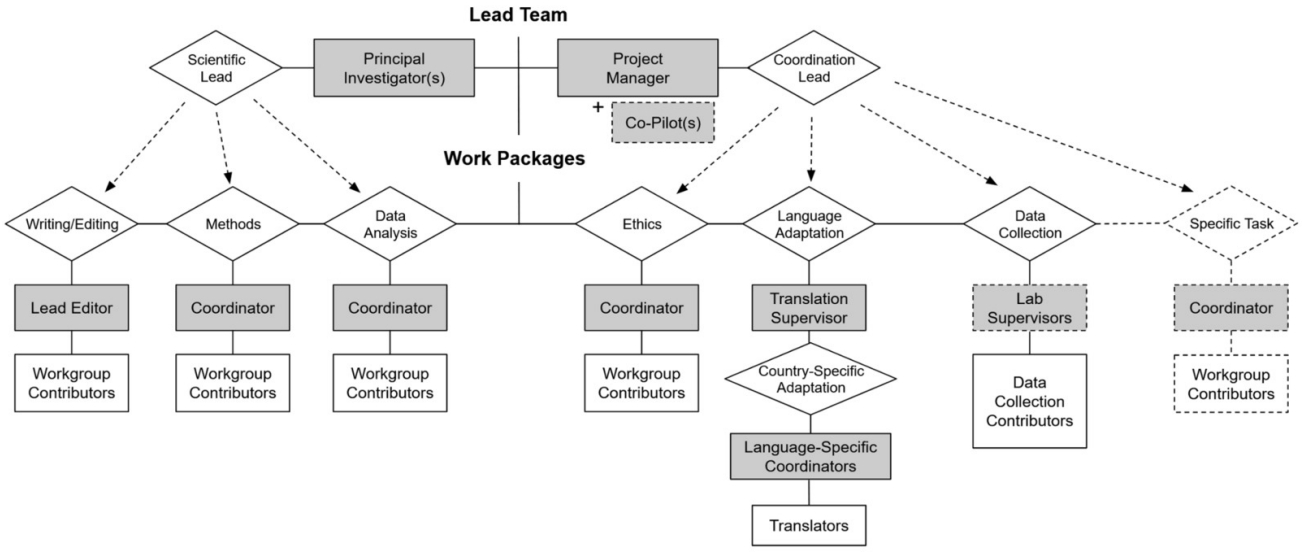


Figure 4. Example of Organizational Structure for BTS

Note. This diagram illustrates an example of organization and relationships between the project lead team—composed of Principal Investigator(s) (PIs) responsible for the scientific content and Project Manager(s) (PMs) responsible for coordination (CRediT: *Project Administration*)—and the various collaborators involved. Diamonds represent the main tasks, organized into work packages. Key responsibility roles are shown in dark rectangles. Dashed shapes indicate optional roles, such as co-pilots, who may be associated with coordinators to support the project's objectives. This example is designed to be adaptable to many BTS projects. The workgroups focus on transversal tasks that support the overall project. These include coordination of work packages such as Writing, Methods, and Data Analysis, with specific responsibilities assigned to designated coordinators and contributors. The diagram also highlights the roles of language-specific coordinators, lab supervisors, and translators, along with the collaborative efforts of participating labs. Each work package and its coordination can be linked to relevant contributor roles using, for example, the CRediT taxonomy (https://credit.niso.org/).

for the lead team, and must be finalized early in the collaboration.

Many scientific projects rely on informal social contracts to define roles and responsibilities. However, as BTS projects scale up, the likelihood of issues, such as authorship attribution, also increases (e.g., Smith et al., 2020). To address these challenges, formalizing various aspects of the collaboration, including a formal contract (e.g., policy document or collaborator agreement; Moshontz et al., 2021), is essential. A clear communication policy can prevent many issues, particularly by establishing a single point of contact for each workgroup and lab. This document should clearly define roles and individual responsibilities (e.g., project manager, coordinator), ensure collaborators are informed in advance, and may include a project philosophy, clear goals, and the decision-making process (see Table 2, also

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for a Consideration List and examples, see supplementary materials at https://osf.io/fb8gt/). Additionally, it can outline onboarding procedures and specify how and by whom data quality will be monitored throughout the research process. By clarifying expectations and specifying consequences for collaborators who fail to meet their commitments, such a communication policy establishes formal rules and helps reduce the risk of misunderstandings.

Fairness but not Fuzziness with the Contributors' Lack of Involvement

A common collaboration challenge in BTS is managing varying levels of collaborator involvement and interest, where some researchers may prioritize speed over data quality or lack long-term commitment. To address this, the lead team should regularly remind collaborators of the importance of high research integrity and the long-term impact of their contributions. The project could be divided into work packages of varying complexity and involvement levels, allowing contributors to align their capacity. Crucially, authorship placement should be explicitly linked to the level of involvement, with those making more significant contributions positioned closer to the first or senior/corresponding author[3]. It is imperative to articulate the potential consequences for collaborators who fail to meet these standards, including removal from the authorship list.

Before commencing work, a co-authorship statement should be prepared, explicitly outlining the criteria for inclusion as a co-author and any limitations on the number of co-authors per project or lab. Furthermore, addressing contribution assessment and self-crediting requires clear guidelines to ensure fair recognition of each collaborator's work, leveraging established frameworks such as CRediT (Brand et al., 2015) or MeRIT (Nakagawa et al., 2023). Crucially, though, it is important to emphasize that authorship is not merely a privilege or right but a responsibility. By accepting authorship, individuals endorse the work as it has been written and executed, standing by it irrespective of any future developments or debates (e.g., harm it may have caused the community; Moffatt, 2011).

Covering Lead Team Transition

The greatest risk to a BTS project occurs when the lead team steps away, potentially leaving the project without guidance or access to crucial documents and data. Such departures, due to career changes, time constraints, funding issues, or personal reasons, can hinder progress, disrupt communication among collaborators, and even jeopardize the project, resulting in a significant loss of human effort and resources. It is crucial to inform collaborators promptly if the lead team is considering leaving or has already left. However, every effort should be made to find a replacement before declaring the project abandoned, including reaching out to experts within and outside the project network.

To mitigate these risks and ensure project sustainability, the lead team must, right from the outset, establish clear policies regarding data ownership, data management, and procedures for handling the departure of essential team members. Whenever possible, responsibilities should be distributed among multiple lead team members through a formal collaboration agreement to guarantee project continuity. Regular meetings among the lead team are essential for maintaining momentum and providing a structured communication model. Additionally, implementing effective project management tools and shared platforms streamlines workflows, enhances collaboration, and ensures transparency. For example, using a generic project-dedicated email with shared access guarantees continuous information flow even if a lead team member leaves or cannot fulfill their duties.

Capacity Challenges: Coping with Human Limitations

Another challenge is workload management and task tracking, which can often exceed individual cognitive, physical, and emotional capacities. This section discusses the importance of logistical support and procedures to address these limitations (Table 3).

Enhancing Collective Writing Productivity

Most journals require contributors to write or edit the manuscript to be considered authors (Coles et al., 2023). Online collaborative writing tools make it possible for everyone involved in BTS projects to suggest edits, contribute ideas, and ultimately earn authorship. However, even though the lead team can accept or reject these suggestions and track contributors, managing input from so many collaborators is inherently energy- and time-consuming. Furthermore, excessive editing rounds from numerous authors can compromise coherence and quality, resulting in cluttered and hard-to-follow text. Additionally, the scope and depth of contributors' comments vary wildly: Skilled contributors may identify significant logical and conceptual flaws, substantially improving the paper, while others may offer less substantive feedback, such as minor grammatical corrections or repetition of previously addressed points.

To combat these limitations, we suggest nominating an editing workgroup, including a lead editor responsible for final editing decisions. The lead editor(s) should communicate expectations regarding the type and nature of acceptable edits, as well as strongly encourage collaborators to provide solutions along with their feedback, emphasizing

3 Please note that it is also important to consider the relevance of such places based on the countries or institutions of the collaborators to ensure fair and efficient decision-making (see also Diversity Challenges).

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Table 3. Summary of Recommendations for Capacity Challenges

Challenges	Recommendations
Enhancing Collective Writing Productivity	• Nominate a lead editor • Allocate time to review drafts and incorporate feedback • Save multiple versions to track changes and contributions, avoid documents being cluttered with comments • List acceptable edit types and provide guidance on the level of feedback expected (e.g., general or detailed).
Tracking, Transparency, and Logistical Organization	• Implement consistent data formats and file naming conventions early on • Maintain a comprehensive data dictionary defining variables, units, and accepted values • Adopt a file organization system and use version-controlled platforms for centralized documentation, tracking changes, and managing contributions • Ensure methodological compliance and track deviations from the finalized protocol.
Benefits and Limits of Artificial Intelligence Models	• Be aware that AI can introduce errors, biases, disengagement, ethical concerns, budget constraints, and additional costs (e.g., data quality issues, subscription fees) • Use AI as a supportive tool under full human oversight

reasonable and constructive criticisms. Further, the editing workgroup must communicate a reasonable timeline for editing and reviewing the developing drafts. When creating this timeline, factors such as different time zones and accommodating collaborators' other commitments should be considered. The editing workgroup must balance feedback with the project's timeline: more collaborators providing deeper, constructive feedback can improve the manuscript, but also extend the length to integrate their feedback. Generally, we recommend overestimating the required time and erring on the side of caution.

To achieve an efficient editing process, creating backups and multiple copies of the draft is advisable. Contributors could be assigned individual copies, or access to comment and provide feedback could be limited to smaller subgroups. In the most comprehensive application, this could be one version per contributor. While this increases the number of documents to track, software solutions exist that can compare and merge multiple documents. This approach can reduce the volume of direct comments on a single comment and mitigate confirmation bias, which often leads people to focus on previously identified elements. Ultimately, by reducing distractions for collaborators reviewing the paper, this could help highlight important flaws or issues and serve as an effective way to identify and assess each contributor's input fairly. Alternatively, collaborators could be asked to provide a general review instead of commenting throughout the document. This approach reduces the number of redundant comments and more readily captures the overall limitations of the paper. More detailed commenting or editing access can then be selectively granted to contributors who request it.

Tracking, Transparency, and Logistical Organization

Managing the overwhelming amount of information required to track collaborators, progress, and document an entire project is both challenging and time-consuming, potentially leading to mistakes or personal exhaustion (see also Volunteering challenges). To ensure project quality, it is crucial to address methodological concerns and deviations from the finalized method. While data collection typically begins after several preparation steps (e.g., methods, ethics), changes may still occur during this process. Insufficient organization or unclear instructions can result in inconsistent implementation and reduced study quality. Step-by-step documentation, clear communication, and robust file management are therefore essential for guiding contributors, reducing uncertainty, and improving efficiency.

Additionally, the lead team is responsible for tracking and documenting both collaborators' and the project's progress. For example, to ensure fair treatment, it is ideal to track each contribution to determine author ranking (see Kohrs et al., 2025 for further discussion on contributorship). To reduce the burden on the lead team, a system that allows individuals to self-assess their involvement is often preferred (Holcombe et al., 2020).

More globally, to facilitate tracking and transparency, we suggest using dedicated project management software, though most are not fully adapted to BTS[4]. Alternatively, shared tables are the most commonly used tools in many projects to organize information and be publicly accessible for greater transparency with all contributors.

Inconsistent data management is challenging for BTS research. The fragmented nature of data presentation, as highlighted by Baumgartner et al. (2023) who identified 69

4 There are specific business software applications to track and organize projects, each with its pros and cons, but these are not specifically designed for BTS projects. Alternatively, STAPLE is a free-software dedicated to BTS projects, created by and for researchers (https://staple.science/).

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disparate data formats, often leads to misunderstandings and renders valuable data unusable, incurring significant resource waste. There are no established international solutions to address this pervasive BTS issue. However, one recommended approach is to adhere to a clear file organization structure and naming conventions, with selective access permissions and implement robust, preplanned and scripted data validation procedures to reduce tedious, error-prone and time-consuming manual validation. The data management workgroup could use pilot or simulated raw data to develop analysis protocols for their data management plans. This approach would enable the verification and proper formatting of data files before uploading them to project servers. The development of data management and analysis plans can be further improved by incorporating a data dictionary (a "codebook") with clear descriptions, easy-to-understand names, and an agreement on data structure formats. Adopting such standardization rules across all participating labs can ensure consistency and prevent costly errors, thereby enhancing data quality, reliability, and reproducibility (see Baumgartner et al., 2023 for further discussion).

Benefits and Limits of Artificial Intelligence Models

The capacity challenges in human-led projects resonate with the rise of generative artificial intelligence (AI) interfaces and Large Language Models (LLM). Consequently, BTS researchers might consider relying on AI assistance. While these technologies are constantly progressing and offer benefits, their use also raises quality and ethical concerns that require careful evaluation.

Most business project management software incorporates AI features that suggest time predictions and resource monitoring based on historical data. While such tools can be useful for tracking progress and may help alleviate the cognitive load on BTS project managers, their adaptability to the specific context of BTS projects, which involve multiple labs and unique needs, is often limited. In addition, because most subscription-based services scale with team size, their costs may become substantial for BTS projects operating on limited budgets or prioritizing resources for the inclusion of underrepresented countries.

Beyond these practical and financial limitations, the use of AI-enabled project management tools also raises several important ethical and regulatory concerns. Data privacy and compliance with local regulations are particularly salient, as cloud-based systems may store sensitive participant data on servers located in other countries, potentially violating local privacy laws. Some platforms may automatically share project files with third parties, creating additional confidentiality risks. Moreover, AI-generated summaries or automated reports could inadvertently expose identifiable information.

More generally, AI tools are increasingly used to support tasks such as literature review, information synthesis, citation tracking, and question answering. These tools leverage machine learning to help researchers navigate the ever-growing volume of academic databases, summarize findings, and identify relevant work more efficiently. While their capabilities are rapidly advancing, they also raise important considerations around accuracy, transparency, and responsible integration into scientific workflows (Knöchel et al., 2025). AI tools create a unique risk where they may group studies together based on shared vocabulary while ignoring critical methodological or theoretical nuances. For instance, an AI tool might synthesize a 'consensus' on a BTS topic by grouping high-powered longitudinal studies with low-fidelity simulations, failing to flag that their underlying data quality is fundamentally different. This differs from traditional search limitations because the AI presents a polished, authoritative narrative that can mask these scientific discrepancies from the researcher. Additionally, AI tools often prioritize the most cited papers—typically those in English, appearing in flagship journals, and indexed in databases affiliated with the software providers. This potentially overlooks the niche, culturally diverse or orphan data that BTS projects could be interested in. In contrast, the collaborative nature of BTS, which leverages the collective expertise of numerous researchers, enables the identification of significant but underrecognized studies that warrant rigorous scientific consideration and offer unique perspectives. Consequently, expert crowdsourcing appears to be a more appropriate and effective strategy for capturing relevant literature than current AI-based approaches.

For BTS, perhaps the most significant advantage of AI-based tools lies in their language translation capabilities, enabled by LLM. These tools are likely to reduce the workload for teams and help bridge language barriers, facilitating clearer and more accessible communication (Knöchel et al., 2025). LLMs can translate or transcribe direct communication and generate preliminary translations within minutes. For instance, video conferencing platforms offer built-in transcription services that provide real-time subtitles, which are becoming increasingly accurate and are likely to offer high-quality translations in the near future—enhancing inclusivity in multilingual collaborations. Other services are also capable of analyzing meeting content to generate minutes, extract key decisions, and identify team deliverables. Thus, these tools have the potential to improve accessibility during meetings and enhance understanding of required actions and decisions afterward. It is crucial, however, to emphasize the human responsibility to ensure the accuracy, ethical compliance, and overall appropriateness of the output at each step.

Finally, AI tools raise important questions about the role of human involvement. As highlighted earlier, sustained human effort is essential to the success of rigorous BTS projects. Over-reliance on automation risks disengaging collaborators and may lead to significant issues related to data quality, authorship, and accountability. When a project relies heavily on automated tools, collaborators may be excluded from meaningful participation in many tasks, which undermines the core principles of BTS: the collaborative aspect built on sharing expertise. This loss of engagement deprives projects of critical human judgment, particularly in tasks requiring nuanced interpretation. For collaborators, if restricted solely to data collection, their

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motivation may hinge primarily on authorship, potentially resulting in lower-quality contributions. Furthermore, limited involvement beyond data collection raises legitimate concerns about the validity of their authorship.

Limitations of the Current Study

Although this paper offers practical solutions based on our experience leading Big Team Science projects, it does not fully address the perspectives and experiences of participating labs and contributors in non-leading roles. We chose the present focus deliberately, as we view effective leadership as the most pressing challenge at this stage. Without clear guidance and robust organizational structures, contributors may lack direction and motivation, resulting in inefficient use of time and resources. Examining the viewpoints of contributors in non-leading roles represents an important avenue for future research. Further, while our lessons learned are drawn from experiences with nearly completed or finished BTS projects (i.e., those resulting in a scientific product), it is crucial to acknowledge the potential for survivor bias. We may be less familiar with the factors that can hinder or prevent BTS projects from even starting. Gaining insights into these early-stage challenges will be essential for expanding the reach and inclusivity of future BTS initiatives.

Future of BTS

A key conclusion hinges on our specific definition of BTS. It transcends mere project size or data collection method; it represents a genuine embrace of the underlying philosophy: collaboration and shared expertise. This means that not all large projects are equal. Employing numerous collaborators to collect data, with the primary goal of increasing statistical power, does not qualify as BTS if contributors are merely seen as means to an end. BTS demands a mindset shift that prioritizes collective effort over individual egos. This shift entails moving from outdated academic stereotypes (e.g., the lone genius model; Buchanan & Lewis, 2023) to embracing more liberating forms of organization with a horizontal hierarchy (e.g., Carney & Getz, 2009) that fosters confidence, collective potential, and fairness. This approach aligns well with *Vertical-Shared Leadership Coevolution Theory* (van Knippenberg et al., 2025), which posits that effective leadership emerges through a dynamic interplay between designated leaders (vertical leadership) and distributed leadership within the team (shared leadership). The lead team holds a crucial role in this model, with various facets emphasizing responsibility over power. They are expected to provide an initial vision, serving as a guiding framework for the collective effort, but without needing to be authoritative figures. Instead, they function as motivators and facilitators, organizing the work and endorsing more responsibility than power. If the project fails, they bear that responsibility. Conversely, when it succeeds, the lead team has provided the right contextual framework; yet, the success is shared and attributable to the entire BTS collaborative.

While this shared leadership model holds great potential, it may also lead to motivational loss, power struggles, and reduced decision-making efficiency, affecting team members across all hierarchical levels as well as the overall project outcome (Chen & Zhang, 2023). Therefore, BTS leads must continuously reflect on the effectiveness of their chosen power dynamic and adjust it when necessary. This underscores the importance of adaptability and critical reflection in shared leadership settings like BTS.

Sustaining a Growing BTS Community in a Fragile World

As BTS becomes increasingly global and multicultural, effective collaboration relies on platforms that uphold academic freedom while promoting inclusion, diversity, equity, and accessibility (IDEA). Central to this collaboration is strong network visibility: connecting with like-minded scholars or recruiting participating labs often depends on multiple channels, including reviewing publications on BTS methods and prior collaborative projects (see our curated reading list: https://osf.io/fb8gt/) or attending dedicated events such as the annual BTS conferences (https://bigteamscienceconference.github.io/).

Funding plays a crucial role in enhancing BTS projects by supporting essential components such as organizational infrastructure, software subscriptions, personnel, and participant compensation. In practice, however, BTS initiatives often rely on fragmented funding sourced from individual local teams, creating significant disparities. This decentralized model favours better-funded teams, potentially undermining equity when some collaborators can finance data collection across multiple sites, or leading to the exclusion of under-funded local teams where expertise is highly relevant.

Compounding the issue further, traditional grant proposals often limit the number of principal investigators (PIs) who can officially represent a project. Because BTS collaborations typically involve dozens or even hundreds of researchers, this constraint forces difficult decisions that often prioritize senior researchers from well-established institutions while sidelining valuable contributors. Furthermore, certain local or national funding schemes impose restrictions on international collaboration (e.g., prohibiting work with researchers from countries engaged in conflict or under international sanctions). These administrative constraints can stifle the open, cross-border cooperation that BTS is designed to promote. They risk erroneously equating individual researchers with their governments' policies, even though many scholars strive for an interconnected academic community, independent of political agendas. Because these policies are imposed by funding agencies operating within political and legal frameworks, researchers must adopt a comprehensive approach to critically evaluate them and seek to reshape existing restrictions.

Final Remarks

As BTS initiatives scale up, it is crucial to acknowledge the structural challenges that accompany such growth. In-

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creasing the number of participating labs introduces risks such as threats to fidelity (Ellefson & Oppenheimer, 2023) or procedural quality (Baumeister et al., 2023). Therefore, BTS approaches must avoid becoming a 'super-Mturk,' where participating labs contribute minimally without meaningfully shaping the research process. These issues can lead to resource wastage, diminishing BTS's benefits and amplifying its risks, leading to a "mega-mistake" (Forscher et al., 2023). The scientific community must confront significant methodological and structural threats.

On the methodological side, simply increasing the sample size does not automatically guarantee broader generalizability. There is a fallacy in considering size as the sole solution (Ghai et al., 2024, 2025). To avoid this fallacy, BTS projects must account for population variability by deliberately including data from low- and middle-income countries, thereby overcoming the common limitation of Western-centric research in psychology.

On a broader level, there is a need for a cultural shift in how scientists initiate, conduct, and produce knowledge. Global research must embrace countries with diverse philosophies of science and cultural values for specific constructs and applied research (e.g., poverty, substance abuse, gender expression; Adetula et al., 2022; Blasi et al., 2022; Henrich et al., 2010; Rad et al., 2018). Addressing these challenges will be essential for BTS to realize its full potential as a rigorous and equitable model of collaborative science.

Author Contributions

Conceptualization: DV (lead), JP (supporting), KS (supporting), JW (supporting)

Data Curation: EB (lead), AE (supporting), DDM (supporting), MK (equal), EP (equal), DV (supporting)

Investigation: all authors (equal)

Methodology: DV (lead)

Project administration: DV (lead), EP (supporting)

Supervision: DV (lead), EP (equal), JM (supporting), JP (supporting), KS (supporting), JW (supporting), ME (supporting)

Visualization: EB (lead), DV (equal)

Writing - Original Draft: DV (lead), ME (equal), EP (equal), FM (supporting), JM (supporting), MM (supporting), RM (supporting), CO (supporting), MT (supporting), YFY (supporting)

Writing - Review & Editing: all authors (supporting; leading DV, ME, EP)

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Ethics Statement

This study did not involve testing of human participants.

Competing Interests

The authors declare no competing interests.

Data Availability Statement

All relevant materials used for this paper can be found on the project page at https://osf.io/2apxb/. This notably includes the survey material and the datasets.

Disclosure of AI use

During the final writing process of this paper, the authors used Large Language Models to improve certain aspects of the English writing style and to check for grammar and spelling mistakes. At each step, the authors reviewed and edited the content, and they take full responsibility and authorship for the publication.

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Appendix A. Description for Each Item in Figures 1-3

Figure 1: Diversity Challenges		
Adaptation		
	Ethics Approval	General delays or complications in obtaining project ethics approval.
	Multiple Ethics Boards	Navigating approval from numerous institutional ethics boards.
	Site Differences	Variability in procedures, infrastructure, or participant populations across research sites.
	Translation Automation	Challenges in automating the translation of materials for diverse languages.
	Translation Coordination	Difficulties in managing manual translation processes across different teams.
Culture		
	Cross-Cultural Communication	Misunderstandings or barriers due to language or cultural communication styles.
	Cultural Comparisons	Issues arising from comparing results across cultures with differing norms.
	Funding Challenges	Difficulty securing or distributing funds across project sites.
	Ideological Resistance	Reluctance to collaborate or share data due to competing ideologies, competition, or fears about idea theft.
	Journal Constraints	Journal limitations (e.g., authorship restrictions or publication delays) that hinder the reporting of work from large collaborative teams.
	Leadership Diversity	Absence of diverse perspectives or representations in leadership roles, which can hinder inclusivity and effective decision-making.
	Project Ideology	Conflicts over the project's overarching goals, theoretical approach, or core values.
Logistics		
	Collaboration Agreement	Problems negotiating agreements on authorship, data sharing, or responsibilities.
	Collaborator Availability	Team members have limited time or inconsistent availability.
	Onboarding Process	Inefficiencies in training and integrating new collaborators into the project.
	Response Delay	Delays in communication or feedback that slow decision-making and overall project progress.
	Time Zones	Coordination difficulties with team members in different time zones.
	Sample Representation	Challenges ensuring participant samples are diverse and representative.

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Figure 2: Volunteering Challenges		
Distribution		
	Diffusion of Responsibility	Tasks falling through due to everyone assuming someone else will do them.
	Leadership Team Size	Problems arising from too large or too small leadership teams.
	Organization Structure	Challenges with defining or maintaining effective team structures.
	Power Distribution	Establishing transparent leadership and defined roles to guarantee fair and efficient decision-making processes.
	Power Imbalances	Problems coming from unequal distribution of power and conflict resolution.
	Project Management	Difficulties in coordinating tasks, timelines, and responsibilities.
	Task Ownership	Uncertainty or disputes over who is responsible for specific tasks.
Interpersonal Relations		
	Emotional Labor	The emotional toll of maintaining the lead team's morale and motivation.
	Interpersonal Issues	Conflicts, miscommunications, or strained relationships between team members.
	Lack of Trust	Selecting trustworthy collaborators with clear criteria and unbiased decisions.
Sustainability		
	Contributor Turnover	A high rate of collaborators leaving the project.
	Lead Team Abandonment	The leadership team withdrew from the project entirely.
	Lead Team Investment	Perceptions of insufficient effort or priority from the leadership team.
	Meeting Needs	Failure to address the practical, professional, or personal needs of contributors.
	Project Continuation	Concerns about the project's long-term viability and completion.
	Unsure Commitment	Doubts about the dedication of contributors to the project.
Involvement		
	Author Inclusion	Debates over who is included or excluded as an author.
	Author Non-Contribution	Individuals listed as authors without meaningful contributions.
	Author Self-Credit	Bias and inconsistencies when team members self-credit their contributions.
	Contribution Acknowledgment	Inappropriately acknowledge different levels of contribution (e.g., Tiers).
	Contributor Involvement	Low engagement or limited participation from team members.
	Lab Expectations	Balancing the expectations of participating labs with the project timeline and available resources.
	Lack of Interest	Team members not being motivated or invested in the project.
	No Acknowledgment	Lack of recognition of the collaborators' contributions to the project's success.
	Non-Intellectual Contribution	Questioning authorship eligibility for those whose contributions are limited to logistical or technical tasks.
	Skill Sets	Gaps in necessary skills among contributors.
	Speed Over Quality	Prioritizing rapid completion at the cost of research quality.
	Task Allocation	Inefficient or unfair distribution of work.
	Task Assignment	Problems with how tasks are assigned to team members.
	Work Expectations	Mismatched assumptions about effort or performance standards.

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Figure 3: Capacity Challenges		
Quality		
	Data Storage	Difficulty storing, organizing, or accessing project data securely and efficiently.
	Specific Software	Problems that arise from needing particular software that is costly, unavailable, or incompatible across teams.
Time		
	Managing Comments	Difficulties organizing and responding to feedback in shared documents or platforms.
	Missed Deadlines	Collaborators consistently fail to meet project deadlines.
	Overambitious Timelines	Setting unrealistic timeframes for tasks or project milestones.
	Time Planning	Ineffective scheduling and inadequate estimation of time for project tasks.
Tracking		
	Data Management	Challenges with organizing, cleaning, or maintaining data quality.
	Document Repository	Problems setting up and maintaining a shared document repository.
	Handling Labs	Issues coordinating work among participating labs.
	Management Tools	Challenges with effectively using or implementing project management tools.
	Resource Dependency	Over-reliance on specific people or resources, leading to bottlenecks.
	Tracking Changes	Difficulty tracking changes to document or data from multiple contributors.
	Tracking Collaborators	Challenges with tracking collaborators, roles, progress and contributions.

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